5

DEVELOPING THE SAFETY REQUIREMENT SPECIFICATIONS

Chapter Highlights

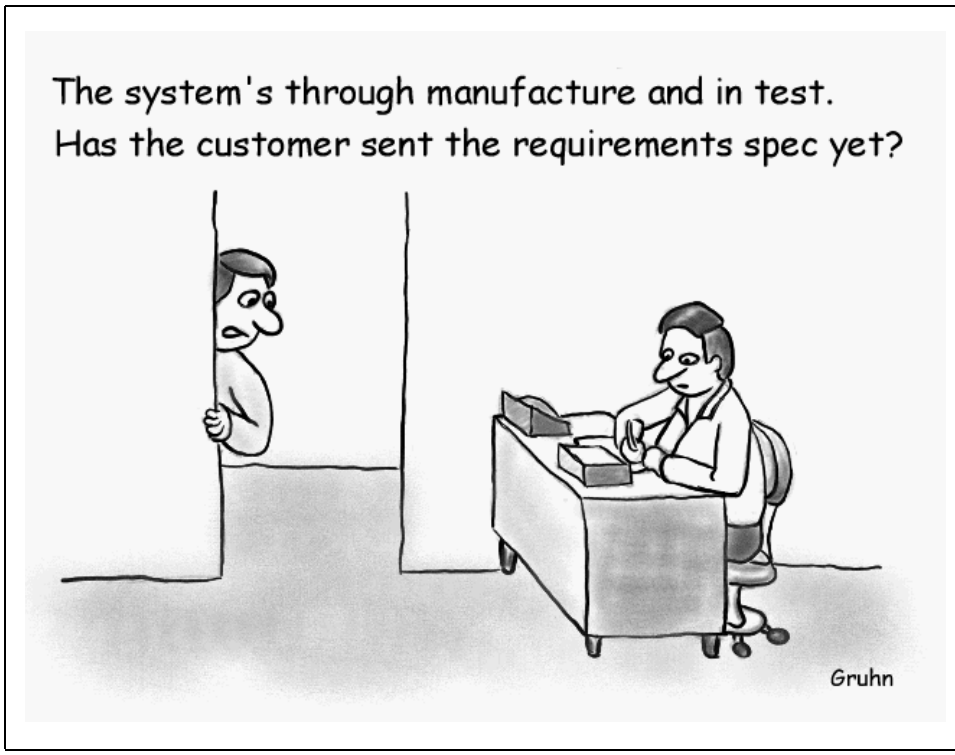
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Nothing can go wrong (click), go wrong (click), go wrong (click)...

5.1 Introduction

Once the need for a safety instrumented system (SIS) is identified and the target safety integrity levels (SIL) have been established for each safety instrumented function (SIF), the safety requirement specification (SRS) needs to be developed. The SRS can either be a single document or a series of documents. The SRS lists the requirements for all functions to be performed by the system. This should also include the application software requirements. The SRS consists of two main parts:

1. The functional requirement specifications
2. The integrity requirement specifications



The functional requirements describe the *logic* of the system. In other words, it defines *what* each safety function should do. For example, low pressure in vessel A will cause valve B to close. The integrity requirements describe the *performance* needed for each function. In other words, it defines the *capability* or *how well* each safety function should perform. For example, the probability or likelihood that valve B will close when the pressure in vessel A is low needs to be greater than 99%. When the term "specification" is used hereafter in this chapter, it refers to both the functional requirement and the integrity requirement specifications.

It's best to design plants that are inherently safe, ones where hazards won't even exist. Unfortunately, this is not always possible. There are times when the only way to prevent a hazard is to add instrumentation that will monitor the process conditions and operate devices to take necessary actions. Deciding what process parameters are to be monitored, what the safety actions should be, and how well the system needs to operate, are all reflected in the functional and integrity specifications.

Chapter 2 noted that the English Health and Safety Executive examined 34 accidents that were the direct result of control and safety system failures in a variety of different industries. Their findings, which are summarized in Figure 2-1, indicated that 44% of accidents were due to incorrect specifi-

cations (both the functional as well as the integrity specification). The second highest percentage of failures (20%) were due to changes made after commissioning. The importance of ensuring that the system specifications are correct cannot be overemphasized. Poor specifications will result in systematic (design) failures. In other words, the system may do what it was designed to do, but that action may be incorrect. Systematic failures are non-random and usually occur under the same conditions. These failures cannot be adequately quantified and are generally controlled by ensuring that the all lifecycle activities are followed.

This chapter identifies some of the problems associated with specification development, reviews ANSI/ISA-84.00.01-2004 (IEC 61511 Mod) requirements, and includes examples of typical documentation that needs to be developed.

5.2 Why Are 44% of Accidents Caused by Incorrect Specifications?

In order to understand *why* the majority of accidents were caused by incorrect specifications, it's necessary to focus on the systems and procedures used to *produce* the specifications. As per the life cycle model outlined in Chapter 2, the following activities need to occur *prior* to the preparation of the specification.

1. Conceptual process design
2. Hazard analysis and risk assessment
3. Application of non-SIS layers
4. SIL determination if an SIF is required

These four steps form the foundation of a correct specification. The procedures used to complete these steps within an organization must therefore be effective and adequate. If procedures are non-existent or inadequate, it doesn't matter how much time or effort is expended in the development of system specifications—the fundamental design basis for the system may end up being incorrect.

The purpose of this book is not to define process design procedures or hazard analysis techniques. Numerous publications on process design and hazard analysis are available from the American Institute of Chemical Engineers Center for Chemical Process Safety and other sources. It should be pointed out that identifying and assessing hazards is *not* what makes a process or system safe. It's the information obtained from the hazard analysis, and more importantly, *the actions taken*, that determine system safety.

Simply performing a HAZard and OPerability study (HAZOP) won't make your plant any safer. After all, the study simply produces a report or a piece of paper. Following the recommendations of the study is what makes things safer.

In addition to the systematic and procedural issues just mentioned, the following are additional reasons why such a large percentage of failures have been contributed to incorrect specifications:

- management systems
- procedures
- scheduling of assessment
- participation of key personnel
- responsibilities not well defined
- training and tools
- complexity and unrealistic expectations
- incomplete documentation
- inadequate final review of specification
- unauthorized deviation from specification

5.2.1 Management Systems

A management system (i.e., a system to ensure effectiveness) must be in place for the lifecycle model to be effective. A management system usually consists of the following steps:

1. identifying the activities in the lifecycle model, and the expected results of those activities
2. setting performance standards against which the results will be compared
3. measuring the actual performance
4. evaluating and comparing the actual performance against the set standards
5. taking necessary corrective actions based on the evaluation

Although it may be relatively easy to develop internal standards and procedures, it may be more difficult to develop and implement a management system to ensure full compliance. The root cause of many problems can often be traced back to nonexistent or ineffective management systems. ANSI/ISA-84.00.01-2004, Parts 1-3 (IEC 61511-1 to 3 Mod)

highlights the requirements, importance and methodology for the management of functional safety.

F. Bird and G. Germain's *Practical Loss Control Leadership* is an excellent text on management systems. [Ref. 2] The term *management system*, although widely used, is somewhat of a misnomer. These systems often have little to do with "management." A better term commonly used is *managing systems*.

5.2.2 Procedures

Procedures need to be developed, reviewed, approved, and understood for all activities in order to effectively perform hazard analyses and SIL determinations. For example, are the SIL determination procedures intended to cover personnel safety only, or are they intended to cover other risk receptors (e.g., environment incidents, financial loss, equipment/property damage, business interruption, business liability, company image, lost market share)? This should be clearly reflected in the procedures.

Hazard analysis usually involves techniques such as fault tree analysis (FTA), failure mode and effects analysis (FMEA), hazard and operability studies (HAZOP), and other similar methods. The HAZOP technique is probably the most widely accepted and used method in the process industry.

SIL determination techniques (the topic of Chapter 6) are not as well established. Large organizations usually have the resources to develop the techniques and procedures. Smaller organizations may have to rely on outside consultants for assistance.

5.2.3 Scheduling of Assessment

Safety assessments are sometimes carried out without the method being well defined or understood simply due to the rush to have the activity completed. Those responsible for planning the assessments need to ensure that the process design and necessary documentation have been completed to such an extent that a meaningful safety analysis can be done. The assessment must not be a mere formality simply to satisfy project management schedules or procedures.

The SIL determination should be an extension of hazard review. This ensures that synergies and understandings developed during the safety assessment are extended to the SIL determination.

5.2.4 Participation of Key Personnel in the Review Process

Most units and systems in process plants are far too complex for any one person to be a total expert on them. One person simply can't be a master on the chemistry of a process, the materials used to contain it, the instrumentation used to monitor it, the control systems used to control it, the rotating equipment used to drive it, the electrical systems used to power it, and the safety techniques used to analyze it.

For this very reason, HAZOP and other hazard analysis techniques usually require a multi-discipline *team*. Persons representing the project, process, operations, safety, maintenance, instrumentation, and electrical groups are typically required. These studies analyze what might go wrong in the operation of a plant. If a hazard is identified and considered significant, a decision must be reached on how to prevent or mitigate it.

When proceeding through the different phases of the hazard analyses and SIL determinations, it's important that each phase not be regarded as a distinct activity. In other words, it should not be viewed that upon completion of one particular phase, the results are simply passed to another team over a "solid brick wall."

5.2.5 Responsibilities Not Well Defined

For each of the four activities listed at the beginning of Section 5.2, the individuals with the lead/responsible roles needs be identified and agreed upon. The individuals assigned must clearly understand their roles and ensure that the analyses are completed in accordance with established procedures. Typical lead roles for each of the four activities are as follows:

Activity	Lead role responsibility
Conceptual process design	Process design
Hazard analysis and risk assessment	Safety engineering
Application of non-SIS layers	Process design
SIL determination if an SIF is required	Safety engineering

5.2.6 Training and Tools

The teams involved with the assessments need to be adequately trained and provided with the necessary tools. For example, prior to an assessment it may be necessary that process control and electrical personnel

receive process training, or that process personnel are adequately trained on the type and capabilities of the control systems. This is essential in order to ensure proper dialog and understanding during the review process. The capabilities and training needs of each team member should be identified prior to the analyses.

5.2.7 Complexity and Unrealistic Expectations

Generally speaking, the simpler a design the better. Complex designs are more difficult to understand, visualize, and review. Take as an example one accident described by Trevor Kletz. [Ref. 1]

An accident in one plant involved a highly hazardous ethylene oxide process. Three valves in series were installed, with two bleed valves between them. As if that weren't enough, these five valves were duplicated so the system could be tested while the plant was on-line without interrupting production.

Operation of the system was carried out automatically by a computer. As a result of a software error, the computer momentarily operated the valves incorrectly. The ensuing explosion and fire of three tons of gas caused extensive damage and resulted in lost production for several months. The software error could be attributed to complexity.

If a safety system is required to prevent or mitigate a hazard, requirements should be stated as simply as possible. Team personnel should refrain from establishing solutions that are too complex, as this can lead to unrealistic and unwarranted expectations.

5.2.8 Incomplete Documentation

Process control and/or instrumentation personnel are usually responsible for the preparation of the safety requirement specifications. The method for documenting the specifications needs to be simple enough for all individuals to understand what is to be provided.

Table 5-1 summarizes the information that should be included in the specifications. The "details of requirement" column identifies whether information pertaining to each item should be provided for the particular application. Comments should also be included as necessary. Additional items are also summarized in the design checklist included in Chapter 15.

Table 5-1: Summary information that should be included in the safety requirements specifications

Item	Details of Requirement
Documentation and Input Requirements	
P&IDs	
Cause and effect diagram	
Logic diagrams	
Process data sheets	
Process information (incident cause, dynamics, final elements, etc.) of each potential hazardous event that requires an SIF	
Process common cause failure considerations such as corrosion, plugging, coating, etc.	
Regulatory requirements impacting the SIS	
Other	
Detailed Requirements for Each SIF	
ID number of SIF	
Required SIL for each SIF	
Expected demand rate	
Test interval	
The definition of the safe state of the process, for each of the identified events	
The process inputs and their trip points	
The normal operating range of the process variables and their operating limits	
The process outputs and their actions	
The functional relationship between process inputs and outputs, including logic, math functions, and any required permissives	
Selection of de-energized to trip or energized to trip	
Consideration for manual shutdown	
Action(s) to be taken on loss of energy source(s) to the SIS	
Response time requirement for the SIF to bring the process to a safe state	
Response action to diagnostics and any revealed fault	
Human-machine interface (HMI) requirements	
Reset function(s)	
Requirements for diagnostics to achieve the required SIL	
Requirements for maintenance and testing to achieve the required SIL	
Reliability requirements if spurious trips may be hazardous	
Failure mode of each control valve	
Failure mode of all sensors and transmitters	
Other	

One simple but effective documentation tool is a cause and effect diagram. (See Table 5.2). This diagram shows the input/output relationship for the safety system, trip set points, SIL for each function, and any special design requirements.

Table 5-2: Cause and Effect Diagram

Tag #	Description	SIL	Instrument Range	Trip Value	Units	Open Valve FV-1004	Open Valve XV-1005	Stop Pump P-1007	Close Valve PV-1006	Notes
FALL-1000	Flow to main reactor R-100	2	0-200	100.0	GPM	X		X		1
PAHH-1002	Reactor internal pressure	2	0-800	600	PSI	X	X			1,2
XA-1003	Loss of control power					X	X	X	X	
XA-1004	Loss of instrument air					X	X	X	X	
HA-1005	Manual shutdown					X	X	X	X	

Notes
1. 2 second delay required before closing FV-1004
2. Reset located at valve PV-1006

The specifications should state *what* is to be achieved, not necessarily *how* to achieve it. This allows others the freedom to decide how best to accomplish the goals, and also allows for new knowledge and techniques.

5.2.9 Inadequate Final Review of Specification

The specifications should be reviewed upon completion and approved by all parties involved with the SIL determination. This will ensure understanding and agreement. Those responsible should not assume that everyone involved had the same understanding during the SIL review. This final review will give everyone an opportunity to reflect on the decisions and to clarify any misunderstandings.

5.2.10 Unauthorized Deviation from Specification

Once the specification is approved, no deviations should be made unless formal change procedures are followed. The system and specification may need to be updated if changes are made during the course of the project. (See Chapter 13, "Managing Changes to a System".) While it may be tempting to make certain changes in order to avoid cost or schedule hurdles, no changes should be made without a proper review and approval process.

5.3 ANSI/ISA-84.00.01-2004, Parts 1-3 (IEC 61511 Mod) Requirements

Subclause 10.3 of ANSI/ISA-84.00.01-2004 (IEC 61511 Mod) provides a summary of the SIS safety requirements. Below are the key requirements of the standard. Refer to the standard for additional details.

The requirements shall be sufficient to design the SIS and shall include the following:

- A description of all the SIFs necessary to achieve the required functional safety
- Requirements to identify and take account of common cause failures
- A definition of the safe state of the process for each identified SIF
- A definition of any individually safe process states which, when occurring concurrently, create a separate hazard (for example, overload of emergency storage, multiple relief to flare system)
- The assumed sources of demand and demand rate of each SIF
- Requirements for proof-test intervals
- Response time requirements for the SIF to bring the process to a safe state
- The SIL and mode of operation (demand/continuous) for each SIF
- A description of process measurements and their trip points
- A description of process output actions and the criteria for successful operation (e.g., requirements for tight shut-off valves)

- The functional relationship between process inputs and outputs, including logic, mathematical functions, and any required permissives
- Requirements for manual shutdown
- Requirements relating to energize or de-energize to trip
- Requirements for resetting the SIF after a shutdown
- Maximum allowable spurious trip rate
- Failure modes and desired response of the SIF
- Any specific requirements related to the procedures for starting up and restarting the SIF
- All interfaces between the SIS and any other system (including the BPCS and operators)
- A description of the modes of operation of the plant and identification of the SIFs required to operate within each mode
- The application software safety requirements (listed below)
- Requirements for overrides/inhibits/bypasses including how they will be cleared
- The specification of any action necessary to achieve or maintain a safe state in the event of fault(s) being detected in the SIF
- The mean time to repair which is feasible for the SIF
- Identification of the dangerous combinations of output states of the SIS that need to be avoided
- Identification of the extremes of all environmental conditions that are likely to be encountered by the SIS
- Identification to normal and abnormal modes for both the plant as a whole (e.g., plant startup) and individual plant operational procedures
- Definition of the requirements for any safety instrumented function necessary to survive a major accident event (e.g., the time required for a valve to remain operational in the event of a fire)

Subclause 12.2 of the standard provides requirements for the specification of the application software safety requirements. It's essential that the

application software specifications be consistent with the safety requirements listed above.

- An application software safety requirements specification shall be developed.
- The input to the specification of the software safety requirements for each SIS subsystem shall include:
  - the specified safety requirements of the SIF
  - the requirements resulting from the SIS architecture, and
  - any requirements of safety planning
- The specification of the requirements for application software safety shall be sufficiently detailed to allow the design and implementation to achieve the required safety integrity and to allow an assessment of functional safety to be carried out.
- The application software developer shall review the information in the specification to ensure that the requirements are unambiguous, consistent and understandable.
- The specified requirements for software safety should be expressed and structured in such a way that they are clear, verifiable, testable, modifiable, and traceable.
- The application software safety requirements specification shall provide information allowing proper equipment selection.

5.4 Documenting the Specification Requirements

Since the specification will form the basis of the safety system design, all the required information should be included as a complete package. The following four items are key documents that should also be included in the SRS package.

1. The **process description**. This should include the following:
   - Piping and instrument drawings (P&IDs)
   - Description of the process operation
   - Process control description including control system design philosophy, type of controls, operator interface, alarm management, and historical data logging
   - Special safety regulations including corporate, local, state/provincial or federal requirements

- Reliability, quality, or environmental issues
- List of operational or maintenance issues

2. **Cause and effect diagram**. A cause and effect diagram can be used to document the functional and integrity requirements in a simple manner that is easily understood by all disciplines. See Table 5-2 for an example.
3. **Logic diagrams**. Logic diagrams can be used to supplement cause and effect diagrams for more complex and/or time based functions and sequences that may not easily be described using a cause and effect diagram. Logic diagrams should be produced in conformance with ISA-5.2-1976 (R1992).
4. **Process data sheets**. Process data sheets ensure that the information required for completing instrument specification sheets is well documented.

Summary

The safety requirement specifications (SRS) consists of two parts: the functional requirement specifications and the integrity requirement specifications. It includes both hardware and software requirements. The functional requirements describe the system inputs, outputs, and logic. The integrity requirements describe the performance of each safety function.

The English Health and Safety Executive examined 34 accidents that were the direct result of control and safety system failures in a variety of different industries. Their findings showed that 44% of accidents were due to incorrect and/or incomplete specifications (both the functional specifications as well as the integrity specifications).

The importance of ensuring that the system specifications are correct cannot be overemphasized.

References

1. Kletz, Trevor A. *Computer Control and Human Error.* Gulf Publishing Co., 1995.
2. ANSI/ISA-84.00.01-2004, Parts 1-3 (IEC 61511-1 to 3 Mod). *Functional Safety: Safety Instrumented Systems for the Process Industry Sector.*
3. Bird, Frank E., George L. Germain, and F.E. Bird, Jr. *Practical Loss Control Leadership*. International Loss Control Institute, 1996.

4. *Programmable Electronic Systems in Safety Related Applications, Part 1 - An Introductory Guide.,* U.K. Health & Safety Executive, 1987.